

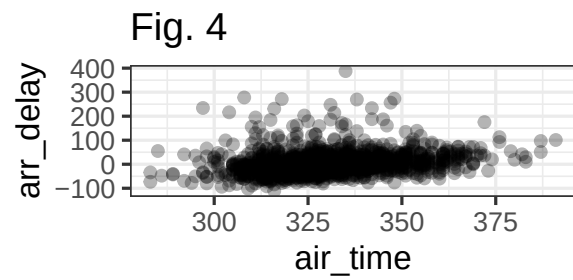
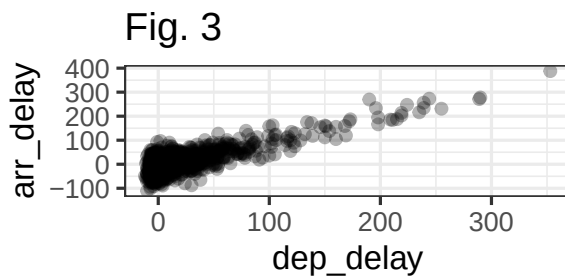
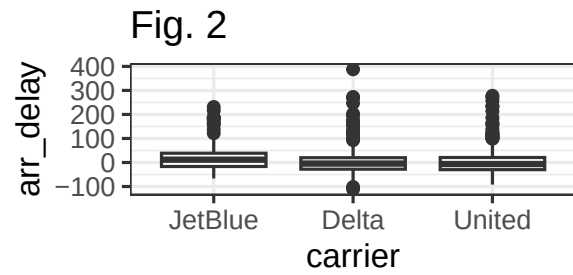
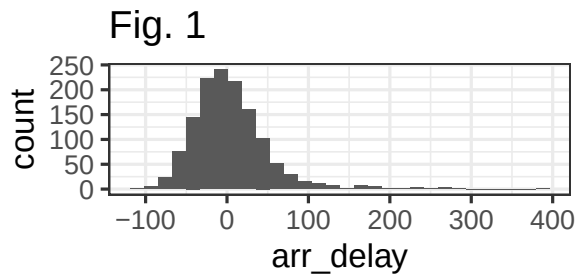
# Multiple Linear Regression: Inference and Model Selection

In this worksheet, we'll practice concepts related to inference for regression coefficients, F-Tests, and model selection. Work through the worksheet carefully in small groups.

## Data: Flights to PDX Airport

We will analyze data on flights arriving to the PDX airport in 2013. We have data on  $n = 1341$  randomly sampled flights, and for each flight, we have 4 variables:

- Response: Arrival Delay in minutes (`arr_delay`). Can be negative if flight arrived early.
- Explanatory 1: Airline Carrier (`carrier`).
- Explanatory 2: Departure Delay in minutes (`dep_delay`). Can be negative if the flight departed early.
- Explanatory 3: Air Time in minutes (`air_time`).



**Question 1:**

Which variables seem to be important predictors of arrival delay? Consider the plots above and context.

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## Question 2:

Let's test if arrival delay varies by airline carrier using ANOVA.

a. Specify the relevant parameters and the set of hypotheses.

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b. State the p-value calculated by R (below) and interpret the result in context. What have you learned about these airline carriers from the ANOVA F-test and Figure 2?

```
mod_carrier <- lm(arr_delay ~ carrier, data)
anova(mod_carrier)
```

Analysis of Variance Table

Response: arr\_delay

|           | Df   | Sum Sq  | Mean Sq | F value | Pr(>F)        |
|-----------|------|---------|---------|---------|---------------|
| carrier   | 2    | 53671   | 26835.4 | 11.122  | 1.621e-05 *** |
| Residuals | 1338 | 3228486 | 2412.9  |         |               |

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### Question 3:

Suppose we decide to model arrival delay using air time, departure delay, and carrier. We are considering additionally including an interaction term between departure delay and carrier. We will use a Nested F-Test to determine if we should use the “small” or “large” model, as defined below.

- Small Model: Predict Arrival Delay with Air Time, Departure Delay, and Carrier
- Large Model: Predict Arrival Delay with Air Time, Departure Delay, Carrier, and the interaction between Departure Delay and Carrier.

a. How many coefficients are in each of the small and large models? What variables do they represent?

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b. Write down the set of hypotheses for a Nested F-Test to compare the small and large model above.

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c. State the p-value calculated by R (below) and interpret the results of the Nested F-Test in context.

```
mod1 <- lm(arr_delay ~ air_time + dep_delay + carrier, data)
mod2 <- lm(arr_delay ~ air_time + dep_delay * carrier, data)
anova(mod1, mod2)
```

#### Analysis of Variance Table

Model 1: arr\_delay ~ air\_time + dep\_delay + carrier

Model 2: arr\_delay ~ air\_time + dep\_delay \* carrier

|   | Res.Df | RSS    | Df | Sum of Sq | F      | Pr(>F)     |
|---|--------|--------|----|-----------|--------|------------|
| 1 | 1336   | 894920 |    |           |        |            |
| 2 | 1334   | 887993 | 2  | 6927.4    | 5.2034 | 0.00561 ** |

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**Question 4:**

Suppose your colleague proposes using a Nested F-Test to determine compare the two models defined below.

- Model 1: Predict Arrival Delay with Departure Delay,  $\log(\text{Departure Delay})$ , and Carrier
- Model 2: Predict Arrival Delay with Air Time, Carrier, and the interaction between Air Time and Carrier.

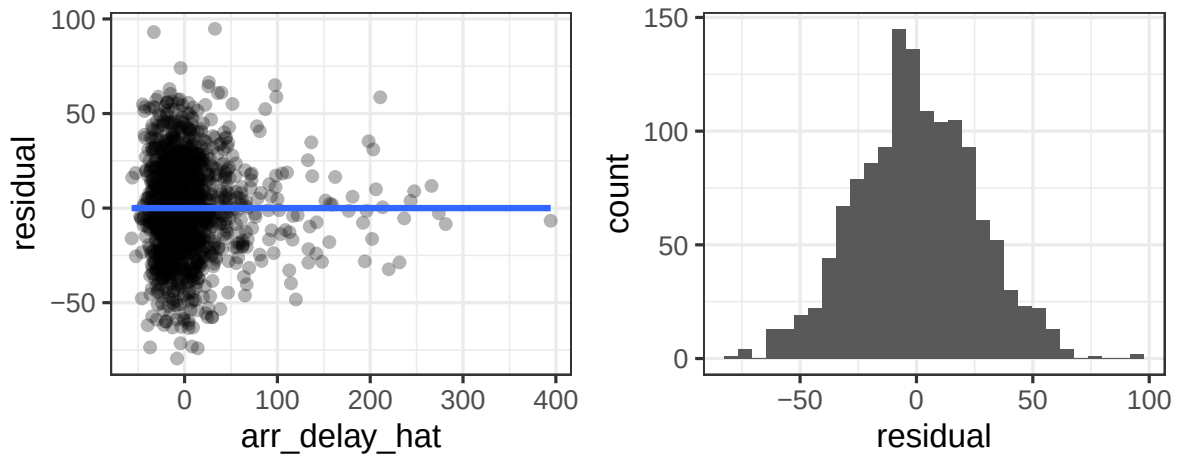
Explain why a nested F-test cannot be conducted to compare these two models.

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**Question 5:**

Below are two plots to help assess assumptions of the large model, as defined before in Q3. Assess the conditions for conducting theory-based inference with linear regression.



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### Question 6:

Examine the regression table below.

```
get_regression_table(mod2, conf.level = 0.8)
```

# A tibble: 7 x 7

| term                      | estimate | std_error | statistic | p_value | lower_ci | upper_ci |
|---------------------------|----------|-----------|-----------|---------|----------|----------|
| <chr>                     | <dbl>    | <dbl>     | <dbl>     | <dbl>   | <dbl>    | <dbl>    |
| 1 intercept               | -295.    | 14.6      | -20.2     | 0       | -314.    | -276.    |
| 2 air_time                | 0.883    | 0.044     | 20.3      | 0       | 0.827    | 0.939    |
| 3 dep_delay               | 0.991    | 0.037     | 27.0      | 0       | 0.944    | 1.04     |
| 4 carrier: Delta          | -9.62    | 2.03      | -4.74     | 0       | -12.2    | -        |
| 7.02                      |          |           |           |         |          |          |
| 5 carrier: United         | -12.8    | 1.97      | -6.46     | 0       | -15.3    | -        |
| 10.2                      |          |           |           |         |          |          |
| 6 dep_delay:carrierDelta  | 0.152    | 0.049     | 3.09      | 0.002   | 0.089    | 0.214    |
| 7 dep_delay:carrierUnited | 0.05     | 0.047     | 1.08      | 0.279   | -0.009   | 0.11     |

a. Write down the estimated model equation implied by the regression table.

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b. Interpret, in context, the coefficients for the (i) intercept, (ii) slope term on air time, (iii) the slope term on departure delay. Be particularly careful for when interpreting (iii) !

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c. Write down the equation for the specific case when the airline carrier is Delta.

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d. What is the confidence level for the confidence intervals shown?

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